Lecture # 3

Four different ways, in which a language can be defined

- So far, we have studied many languages like INTEGER, PRIME, PALINDROME etc.,
- Now, we will take a detailed look at how a language can be defined in different ways.
- There are four ways that we will study in this course:
  - Descriptive way
  - Recursive way
  - Regular Expression
  - Finite Automata

Descriptive Definition of a Language

- The language and its associated conditions are defined in plain English.
- This way is semi-formal way with chances of ambiguity.
- Descriptive method is used in early phases of requirement engineering and resultant equations are then formally transformed using other methods.

- Example:

  The language L of strings of even length, defined over Σ={b}, can be written as

  L={bb, bbbb, …..}
- Example:

  The language L of strings that does not start with a, defined over Σ={a,b,c}, can be written as

  L={b, c, ba, bb, bc, ca, cb, cc, …}

- Example:

  The language L of strings of length 3, defined over Σ={0,1,2}, can be written as

  L={000, 012, 022,101, 101,120,…}

- Example:

  The language L of strings ending in 1, defined over Σ ={0,1}, can be written as

  L={1,001,101,0001,0101,1001,1101,…}

- Example: The language **EQUAL**, of strings with number of a's equal to number of b's, defined over Σ={a,b}, can be written as

  {Λ ,ab,aabb,abab,baba,abba,…}
- Example: The language **EVEN-EVEN**, of strings with even number of a's and even number of b's, defined over Σ={a,b}, can be written as

  {Λ, aa, bb, aaaa,aabb,abab, abba, baab, baba, bbaa, bbbb,…}

- Example: The language **INTEGER**, of strings defined over Σ={-,0,1,2,3,4,5,6,7,8,9}, can be written as

  INTEGER = {…,-2,-1,0,1,2,…}

- Example: The language **EVEN**, of stings defined over Σ={-,0,1,2,3,4,5,6,7,8,9}, can be written as

  EVEN = { …,-4,-2,0,2,4,…}

- Example: The language $\{a^n b^n\}$, of strings defined over $\Sigma=\{a,b\}$, as

  $\{a^n b^n : n=1,2,3,\ldots\}$, can be written as

  {ab, aabb, aaabbb,aaaabbbb,…}

- Example: The language $\{a^n b^n c^n\}$, of strings defined over $\Sigma=\{a,b,c\}$, as

  $\{a^n b^n c^n : n=1,2,3,\ldots\}$, can be written as

  {abc, aabbcc, aaabbbccc,aaaabbbbcccc,…}